Juhyeok **Park**

Senior, B.S. Electronic Engineering, Chungbuk National University · Expected Feb. 2027

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Research interest Perception and verification: checking an AI's decisions against sensor signals before they are executed, and remembering verified experience to decide faster in physical AI

Education

Chungbuk National University *Cheongju, Korea*

B.S. in Electronic Engineering *Mar. 2021 – Feb. 2027 (expected)*

- GPA 3.80 / 4.5 (major 3.84)
- A+ in Probability and Statistics, Random Processes, Communication Theory, Digital Communication, Digital Signal Processing, Signals and Systems, Information and Communication Engineering

Publications · Patent

- **[First author · oral] Juhyeok Park** et al., "An LLM Wiki System for High-Consequence Domain Knowledge Organization: Initial Feasibility for Space Operations and Nuclear Engineering," *3rd IAA Conference on AI for Space*, Jeju, Aug. 2026.
- **[First author · oral] Juhyeok Park** et al., "A VLM robot manipulation pipeline for mitigating physical hallucination" (in Korean), *Korea Society of Industrial Information Systems Spring Conference*, 2026 (corresponding author: Hogeon Seo).
- **[Co-author]** D. Choi, **J. Park**, H. Seo, "Development of an AI agent framework for autonomous improvement of an ultrasonic imaging algorithm" (in Korean), *Journal of the Korean Society for Nondestructive Testing*, 46(4), 268–277, 2026 (KCI; corresponding author: Hogeon Seo).
- **[Co-inventor]** "Directional steering in LLM embedding space with a human in the loop," Korean patent application (pending), 2026.

Research Experience

Korea Atomic Energy Research Institute (KAERI) *Daejeon, Korea*

Research Intern *Mar. 2026 – Aug. 2026*

- Designed and built an LLM wiki knowledge-management system on the nuclear-domain model AtomicGPT: the model writes wiki pages as it works and reads the relevant pages before answering. Retrieval hit rate 55%→79%, answer accuracy 33%→55%. Implemented on internal and external network servers and handed over; now being built into an institute service (IAA 2026, first author)
- Developed a pipeline that checks VLM-generated robot manipulation commands with forward kinematics before execution. Designed controlled experiments over degrees of freedom (1→2→5) and four hint types; collected and analyzed the data and wrote the paper (KSIIS 2026, first author)
- In a framework where an LLM agent repeatedly edits and evaluates ultrasonic SAFT imaging code, designed the evaluation scheme and benchmark tasks (including an extension for an institute-wide service) and documented the code (J. Korean Soc. NDT 2026, co-author)
- Ran the experiments and built the demo for a patent on directional steering in LLM embedding space (co-inventor)

Capstone and graduation thesis: noise-robust object detection in satellite images (advisor: Prof. H. Kim) *Chungbuk National University*

Team Lead · model, training and evaluation code *Nov. 2025 – Nov. 2026*

- Noise training plus DINOv2 → YOLOv8s-seg distillation, re-run on the official iSAID split after I found a leaky public split and a label bug: noise training 0.018 → 0.381 noisy mAP50; centred spatial distillation +0.020 / +0.010 (two seeds); relational KD no gain. Deployed model unchanged (22 MB)
- Built a web demo that compares detections across noise levels

AI Competition on Traffic Accident Analysis from Dashcam Videos (Ministry of the Interior and Safety · National Forensic Service, DACON)

5-person team · Stage 2 (collision time and approach direction) *Sep. 2026*

- Built models that determine the collision time and the other vehicle's approach direction from dashcam video, and built a labeling tool

NAVIS Lab, Chungbuk National University *Cheongju, Korea*

Undergraduate Researcher *Dec. 2025 – Feb. 2026*

- Designed and implemented two compensation algorithms for estimating the arrival time of eLoran signals under sample loss on top of the baseline code, and validated them with 1,000 Monte Carlo runs and measured eLoran signals. SNR improved by 6.78 dB but biased the arrival time; plain accumulation proved the most robust

Coursework · Skills

Courses	Electronic Circuits · Digital System Design · SoC Design · Computer Architecture · AI Systems · AI System Semiconductor Design · Automatic Control
Course projects	PPM modulation/demodulation with hard and soft decisions (MATLAB) · Cortex-M3 secure OTP · AES-128/CBC from scratch · Raspberry Pi real-time vision and voice integration
Languages · Tools	Python, C/C++, MATLAB · PyTorch, Ultralytics YOLO, DINOv2 · LLM/VLM pipelines, built on-premise LLM service servers · Linux, Git, LaTeX

Papers, slides and demos: nonojamjam.github.io